

SOP-CLEAN-007: End-of-Batch Cleaning – Rotary Tablet Press RTP-7

Department: Production / Housekeeping

Revision: 02

Effective Date: 2024-01-15

Regulatory Reference: Schedule M, GMP Guidelines

Purpose

This SOP defines the standardized procedure for end-of-batch cleaning of the Rotary Tablet Press RTP-7 to ensure removal of product residues and prevention of cross-contamination between batches. Proper cleaning is critical to maintain product quality, patient safety, and compliance with regulatory requirements such as Schedule M and GMP guidelines. It also ensures equipment readiness for the next batch and minimizes the risk of mix-ups.

Scope

This procedure applies to all production and housekeeping personnel responsible for cleaning the RTP-7 after completion of each batch, during product changeovers, and before maintenance activities. It is mandatory for all batches regardless of product type, with additional requirements applicable for potent or allergenic compounds as defined in product-specific cleaning validation protocols.

Cleaning Agents

Agent	Concentration	Contact Time	Rinse Required
Purified Water (PW)	N/A	Immediate	Yes
Non-ionic Detergent (e.g., Tween 80 solution)	1-2% v/v	10-15 minutes	Yes
Isopropyl Alcohol (IPA)	70% v/v	5 minutes	No
Sodium Hypochlorite Solution	0.5% w/v	10 minutes	Yes

Prerequisites

Before initiating cleaning, ensure the batch is completed and documented in the Batch Manufacturing Record (BMR). Perform line clearance and obtain QA clearance for cleaning. Shut down the machine using standard operating procedure and apply Lockout-Tagout (LOTO). Remove all remaining product materials from hoppers, feeders, and collection bins. Ensure all required cleaning tools (lint-free cloths, brushes, buckets) are clean and labeled. Verify availability of approved cleaning agents and ensure they are within expiry. Personnel must be trained and authorized for cleaning operations.

Safety Warnings

- Always wear appropriate PPE including gloves, goggles, and mask while handling cleaning agents.
 - Avoid direct contact with sodium hypochlorite and IPA as they may cause skin and eye irritation.
 - Ensure proper ventilation when using IPA to prevent inhalation hazards.
 - Do not mix cleaning agents as this may cause hazardous chemical reactions.
 - Dispose of cleaning waste as per hazardous waste management procedures.
 - Ensure electrical components are protected from water ingress during wet cleaning.
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Cleaning Procedure

Phase 1 – Dry Clean

Begin by removing all detachable parts such as hopper, feeder assembly, punches, and dies. Use a vacuum cleaner with HEPA filter to remove loose powder from all accessible surfaces including turret, die table, feeder housing, and compression zone. Avoid using compressed air as it may disperse dust into the environment. Use a nylon brush to dislodge stubborn powder deposits from crevices. Collect all waste material in designated containers and label appropriately. Ensure no visible powder residue remains before proceeding to wet cleaning.

Phase 2 – Wet Clean

Prepare a 1-2% non-ionic detergent solution using purified water. Using lint-free cloths and soft brushes, clean all product contact surfaces including hopper interior, feeder paddles, die table, turret surface, and punch guides. Pay special attention to corners, joints, and threaded areas where residue may accumulate. Clean non-contact external surfaces such as machine body, control panels (using damp cloth only), and guards. For areas with potential microbial risk or potent product residue, apply 0.5% sodium hypochlorite solution and allow a contact time of 10 minutes. Ensure uniform application and avoid pooling of liquid.

Phase 3 – Rinse

Rinse all cleaned parts and surfaces thoroughly with purified water to remove detergent and chemical residues. Use low-pressure water to avoid splashing and ensure controlled cleaning. For critical product contact parts, perform final rinse using fresh purified water. Verify that no foam or visible residue remains. Collect rinse water appropriately and dispose of as per SOP. Avoid excessive water usage near electrical components; use damp cloth wiping where necessary.

Phase 4 – Drying and Inspection

Allow all cleaned parts to air dry in a designated clean area or use lint-free cloths for manual drying. Ensure no moisture remains, especially in crevices and अंदर bore areas, as residual moisture can promote microbial growth. Once dry, visually inspect all surfaces under adequate lighting. Surfaces must appear clean, dry, and free from stains, residue, or odor. Reassemble machine components only after QA or supervisor verification if required. Wipe critical surfaces with 70% IPA as a final sanitization step where applicable.

Acceptance Criteria

All equipment surfaces must be visually clean with no visible residue, stains, or particulate matter. Swab tests may be conducted as per cleaning validation protocol to ensure residue levels are below acceptable limits. There should be no detectable odor from previous product. Equipment must pass QA inspection and be labeled as "Cleaned" with status tag before reuse.

Documentation

All cleaning activities must be recorded in the Batch Manufacturing Record and equipment logbook. Record details including date and time of cleaning, names and signatures of personnel involved, cleaning agents used (with batch numbers), and inspection results. Any deviations or abnormalities observed during cleaning must be documented and reported to QA. Cleaning status labels must be affixed to the machine indicating "Cleaned" and "Ready for Use."

Frequency

This SOP must be followed after every batch completion and during product changeover. In addition, periodic deep cleaning SOPs (e.g., weekly or monthly) must be followed for thorough dismantling and cleaning of internal components. Additional cleaning may be required in case of spills, contamination, or QA instructions.
